Using WordNet to identify language variation within SNOMED CT.

Possible implications for principles of content development, with particular reference to Mental Health.

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Introduction: Background and motivation

SNOMED CT's editorial guide [1] states that:

> *"…content in SNOMED CT must represent unambiguous, clinically relevant information which can be exchanged and understood internationally..."*

This goal assumes that ambiguous terms and phrases have no place in a useful clinical terminology and that elimination of potentially ambiguous content is always desirable.

This ePoster is motivated by the question of the extent to which this editorial ambition hold for all areas of SNOMED CT, and asks: *are there parts of SNOMED CT where clinically valuable words and phrases exhibiting ambiguity appear more frequently? If so, should such content be tolerated?*

Recent efforts to improve current SNOMED CT Mental Health [2] content have led to many difficult to resolve discussions attempting to distinguish between similar words and phrases, for example between:

- 'abnormal affect' and 'disturbance in affect'
- 'anxiety', 'fear ', 'worry' and 'phobia', and
- 'perception' and 'sensation'

This ePoster reports on work testing the hypothesis that Mental Health discourse is reliant, in part, upon words from 'non-specialist language' *more than other clinical domains* and that, in turn, these same words exhibit *more ambiguity* than 'non-specialist language' employed in other clinical domains.

The work performed produces qualitative evidence that may support both aspects of the hypothesis. Following an explanation of the approach taken and presentation of the results, the ePoster goes on to discuss how the results might be used to modify SNOMED CT's general editorial approach in areas where there may be value in tolerating ambiguity.

WordNet as a source of non-specialist language

References in the preceding section to 'non-specialist language' suggest that it can be easily distinguished from 'specialist language'. This is not the case (the boundary is blurred) however a reasonable approximation might be that if a word of phrase is found in a general-purpose dictionary it can *more likely* be regarded as 'non-specialist'.

Many dictionaries, in many languages, are available [3]. This study has chosen WordNet [4] for the following reasons:

1. It is available in an English language form
2. It is of a comparable size to other English language dictionaries
3. Its design places special emphasis on semantic relations between the words it contains (WordNet provides sets of cognitive synonyms - 'synsets')
4. Its design makes it easy to implement directly in computer systems.

Point 3 is particularly valuable when trying to investigate the second part of the hypothesis (the extent of ambiguity), in that WordNet provides an objective measure of semantic similarity between words tested.

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Methods

Data tested

The study was performed using the April 2024 International SNOMED CT data, with comparison made against the content of WordNet 3.0 (English language). Analysis was performed in a python development environment [5], accessing WordNet using features of the NLTK library [6].

Data sampling and preparation

Samples of concepts across (and within) multiple top-level chapters were generated. Details of the actual samples and sizes are available [7], but in summary, 31 samples were tested, taken from all top-level chapters of SNOMED CT except the 'metadata' and 'special concept' chapters, with an average sample size of 3315 concepts (max. 8787, min 172).

Comparison between SNOMED CT and WordNet

Terms from the codes in each sample were broken into their individual words, and the words that remained after stop word removal were tested against the content of WordNet to determine the proportion matched (see Figure 1).

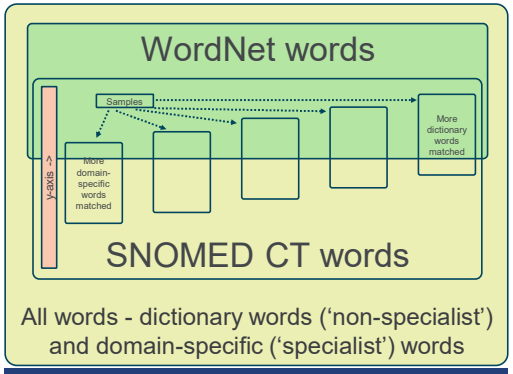
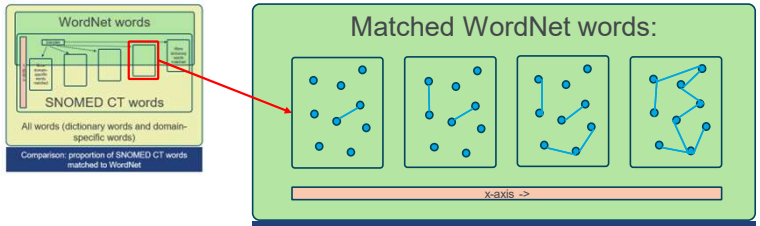


Fig. 1: Comparison: proportion of SNOMED CT words matched to WordNet



Ambiguity: testing for word similarity within WordNet

For each sample, the WordNet-matched tokens (identified according to their inclusion in one or more 'synsets') were compared with one another to give a measure of how semantically 'similar' or 'connected' these words were (or might be).

The python NLTK library offers several similarity measures based on WordNet 3.0. For this work the **path_similarity()** [8] function was used, comparing each 'word in synset' combination with each other. This function returns *"…a score denoting how similar two word senses are, based on the shortest path that connects the senses in the is-a (hypernym/hyponym) taxonomy…",* with the scores based on the degree of separation (1/nodes in path, so 1 for synonymy, 0.5 for direct hypernym/hyponym relation, 0.33 for two steps of hypernym/hyponym relation and so on). This work concentrated on the synonymy and direct hypernym/hyponym returns, using these as a proxy for 'words that may mean more than one thing, or overlap closely with one another' and therefore risk being ambiguous.

As illustrated in Figure 2, testing the words in any given sample of data produces a measure of their overall 'connectedness', from loosely connected (on the left - with the implication that the words have meanings distinct from one another) to more highly connected (on the right – with the implication that the words may have multiple or overlapping meanings in general use).

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Results

Overall results

The full results of testing are available in the code repository [7]. The description below illustrates the nature of the results generated and their synthesis into an overall picture, summarised visually in Figure 3.

As already explained, each data sample underwent two stages of testing, firstly to determine the proportion of words matching those in WordNet and secondly a testing stage *between* the matched words to give a measure of 'connectedness' (and thus risk of semantic overlap, polysemy and ambiguity) within each sample.

WordNet word matching

All samples matched some words to WordNet, varying from as low as 10% ('aerobic bacteria') up to 95% ('physical force', 'event', 'record artifacts and mental health findings'). As illustrated on the y-axis of Figure 3, a large concentration of 'clinical chapter' samples (disorders, findings and procedures) can be seen between 55-85% proportion matched.

Matched word 'connectedness'

A unit-less value for 'connectedness' is calculated for each sample, simply the total number of synonym and hypernym/hyponym associations for each sample divided by the total number of tokens/words. The values generated vary between a low value of 0.01 ('aerobic bacteria') and a high value of 2.13 ('event'). As illustrated on the x-axis of Figure 3, the majority of 'clinical chapter' samples are bunched between 'connectedness' values of 0.3 to 1.2.

Result synthesis

Figure 3 summarises the results of both the 'WordNet matching' and 'connectedness' stages of testing, giving each sample coordinates on the y- and x-axes respectively. Each sample is shown as a small coloured dot with a label indicating its SNOMED CT origin. Where relevant each small dot is surrounded by a larger, paler, disc indicating the proportion of concepts in the sample that are declared sufficiently defined in the SNOMED CT data. Apart from the large blue disc in the lower left of the distribution ('anti-infective products') the majority of samples with high concept definition proportions fall, once more, in the middle section of both the x- and y-distributions (bounded by the green circle). By contrast the samples in the lower left of the distribution (low proportion matched and low connectedness) and upper right (high proportion matched and high connectedness) are largely primitive.

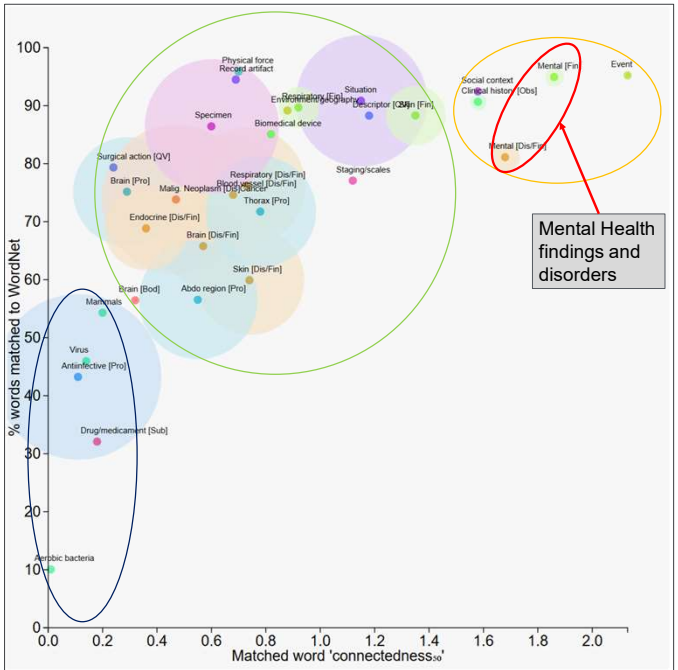


Fig. 3: Summary scatterplot of proportion of words matched to WordNet (y-axis) and 'connectedness' (x-axis). See text for discussion

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Discussion

The ePoster has considered the variation across SNOMED CT with regard to the use of words taken from ‘non-specialist' - as opposed to domain-specific ‘specialist’ technical language.

The results indicate that a significant proportion of words found in terms used in Mental Health discourse are ‘non-specialist’ dictionary words and also highly semantically ‘connected’ (only beaten by ‘events’ – orange and red (mental health) ovals in Figure 3). A sample visualisation of ‘connectedness’ is given in Figure 4.

By contrast the majority of other disorder, finding and procedure concepts show lower matched proportions and ’connectedness’ (green section of Figure 3), and highly domain-specific chapters (e.g. organisms) use far few 'common' words and have almost no ‘connectedness’ (blue section of Figure 3).

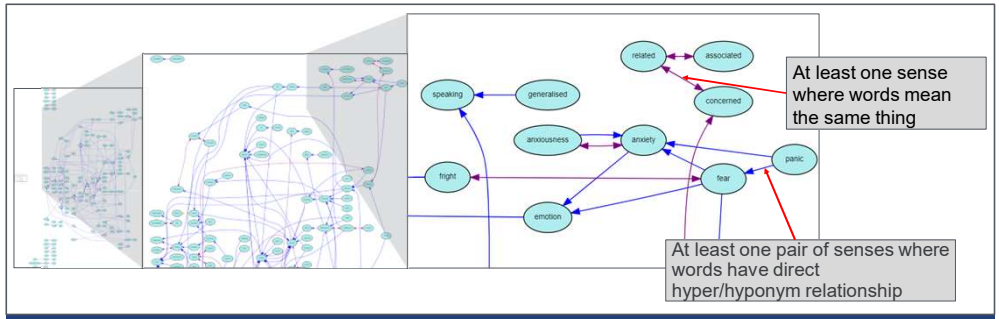


Fig. 4: Zoomed sample visualisation of WordNet ‘connectedness’ calculation – based on words found in terms applied to concepts beneath 48694002 Anxiety (finding)

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Discussion

Whilst essentially a qualitative result, this work highlights Mental Health findings as an outlier and may help make the case for a more inclusive SNOMED CT editorial stance - in particular one that tolerates such content, associating such terms with *multiple possible meanings* and alerts users to the risks, whether this be in the Mental Health chapters or other areas where terms contain a high proportion of words drawn from ‘non-specialist language’.

‘Non-specialist’ and ‘general language’ words remain a valuable element of self-reported Mental Health symptoms, and their exclusion from SNOMED CT on grounds of ambiguity may limit the part it can play in the care of such patients.

Concluding points

It is hoped that this ePoster shines light on the variation of language found in SNOMED CT and stimulates discussion on the possible editorial implications of such an observation.

It is acknowledged that this work has only been performed on an English language version of SNOMED CT. This is a weakness of the methodology but was limited by available tools and motivated by the fact that the questions concerning meaning have arisen in an English language context.

It is also noted that ‘matched word proportions’ and ‘word connectedness’ are unlikely to be independent variables, but it is hoped that this does not detract from the overall message of the study.

References

[1] https://confluence.ihtsdotools.org/display/DOCEG/Scope
[2] https://confluence.ihtsdotools.org/display/MBHCRG/Mental+and+Behavioural+Health+Clinical+Reference+Group
[3] https://en.wikipedia.org/wiki/List_of_dictionaries_by_number_of_words
[4] https://wordnet.princeton.edu/
[5] https://bitbucket.org/cheetham2017/expo2024/src/master/
[6] https://www.nltk.org/
[7] https://bitbucket.org/cheetham2017/expo2024/src/master/results/
[8] https://www.nltk.org/howto/wordnet.html#similarity

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